



COOLER DESIGNS FOR SMARTER PERFORMANCE

ENCLOSURE DESIGN & THERMAL MANAGEMENT OF POWER CONVERSION SYSTEMS

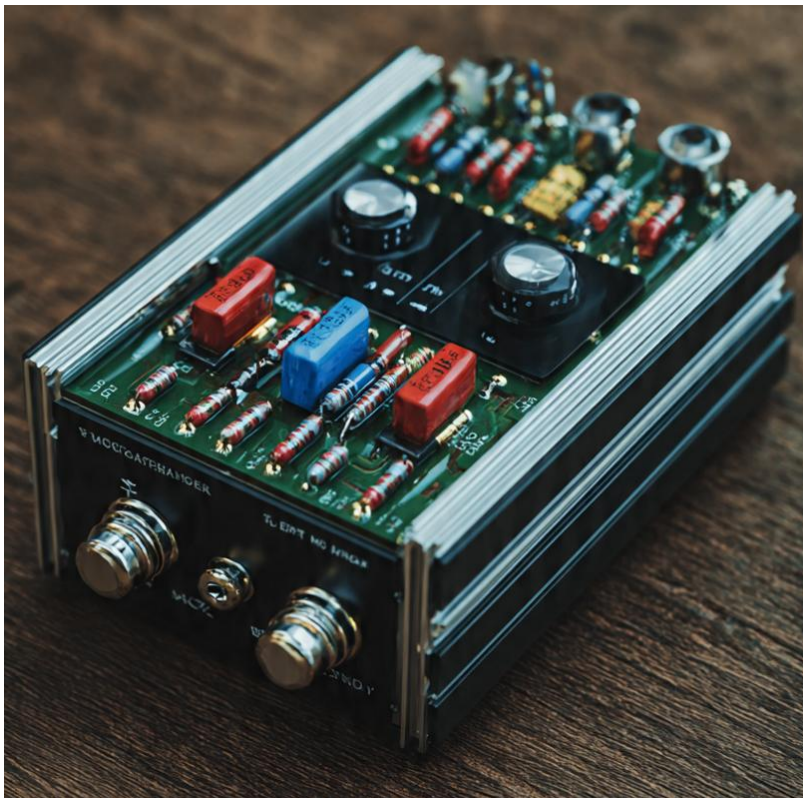


Figure — Generic Power Converter (illustrative)

Shape Your Idea To Reality

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Power Conversion Systems & Our Solutions

Power converters drive the energy behind today's innovations — from electric vehicles to renewable grids. But as designs get smaller and power demands grow, **heat becomes the hidden threat**. Excessive thermal stress can drain efficiency, shorten lifespan, and raise costs.

Our **Thermal Simulation Solutions** help engineers to

- **Predict Hotspots Early** — eliminate surprises before prototyping.
- **Optimize Cooling Solutions** — balance efficiency with cost-effective design.
- **Accelerate Time-to-Market** — validate faster with virtual testing.

Saving time, cost, ensuring reliability and optimize enclosure design, design verification, environmental qualification, and certification.

Our Services

01 · Enclosure Design

- Custom Enclosure Design & Modelling, Material Selection & Optimization
- Aesthetic & Ergonomic Design
- Manufacturability & Assembly Design
- Safety & Compliance Engineering

02 · Cooling Strategy Optimization

Heatsink performance analysis, Enclosure airflow and ventilation studies, Liquid cooling vs. forced-air cooling comparisons, Component derating risk assessment.

03 · Design Optimization

Recommendations to improve board layout, airflow, structural and reliability.

04 · Altitude & Environmental Derating

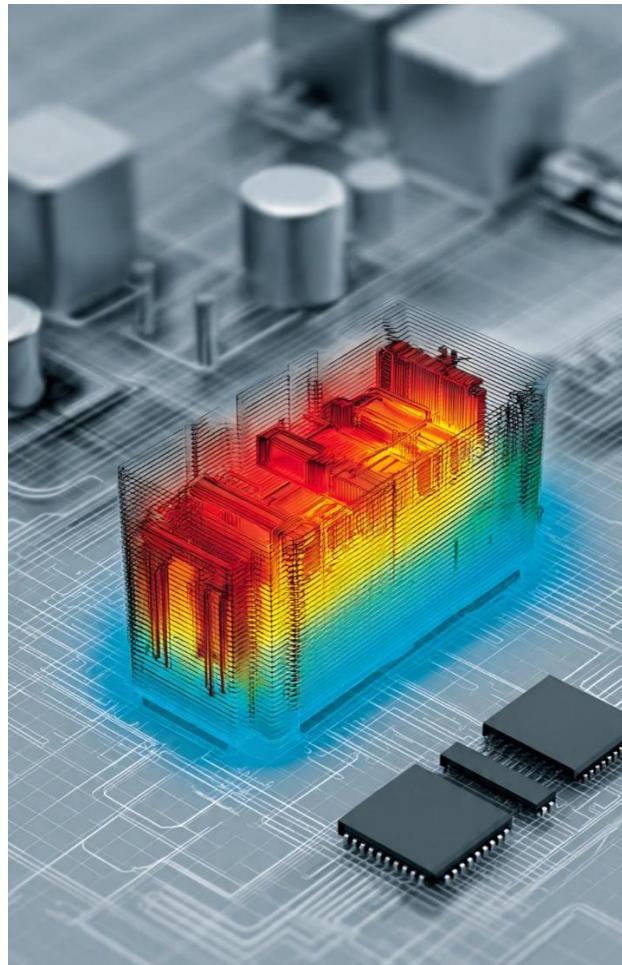
Analysis of reduced convective cooling, pressure, and ambient effects across the specified installation and operating envelope.

05 · Test Correlation

Chamber and bench test support, including thermal imaging and embedded sensors, to correlate and refine thermal models against measured data.

06 · Compliance Reporting

Documentation aligned with industry standards for certification.



Technical Capabilities

STUDY PARAMETERS	SIMULATION OUTPUT
Component-level insights for MOSFETs, IGBTs, and SiC MOSFETs	Junction Temperature, Case Temperature
Heatsink / Baseplate	Baseplate Temperature at rated power and operating ambient
Enclosure Surface	External case/touch temperature at maximum ambient
Operating Temperature Range	Functional range across duty cycle and installation zone
Magnetics, Capacitors	Operating temperature limit as per manufacturer's rating (class B/F/H)
Altitude Derating	Reduced air density impact on convective cooling as per installation altitude (up to 5,000 m)
Power Density	Thermal loading per unit enclosure volume
Thermal Cycling Endurance	Repeated cycling fatigue on solder joints/interfaces

Compliance & Standard Verification

Verification against Industry & Regulatory Standard

- IEC 62368-1 / IEC 60950-1 — Safety of audio/video, IT, and communication technology equipment; thermal and touch-temperature limits.
- UL 508C — Standard for Power Conversion Equipment; overtemperature and thermal protection requirements.
- IEC 60068-2 (Series) — Environmental testing: dry heat, cold, thermal cycling, and damp heat.
- IEC 61800-5-1 — Adjustable speed electrical power drive systems; safety requirements including thermal design.
- MIL-STD-810H — Environmental Engineering Considerations (Method 501.7 High Temp, 502.7 Low Temp, 503.7 Temp Shock).
- IEC 60529 (IP Rating) — Degrees of protection provided by enclosures against dust and moisture ingress.
- IPC-2221 / IPC-9592 — PCB and power conversion equipment design/thermal management standards.
- Manufacturer/OEM Interface Control Document (ICD) — enclosure-specific thermal & mounting requirements.

Deliverables

Thermal Analysis Report (TAR) documenting

Simulation Modelling, Inputs, Assumptions & Boundary conditions

Simulation Results: Heat maps, temperature distribution plots, airflow patterns etc.

Simulation dashboards with performance metrics

Executive summary highlighting risks and recommendations

Design Recommendations

Risk Assessment

Compliance-ready documentation for certification

Engagement Model

Requirement Analysis: Scoping workshop to capture the power loss map, switching profile, duty cycle, and installation/mounting zone etc.

Pre-Design/Simulation Review: Enclosure Design, CFD/FE model build, boundary condition definition, and analysis execution.

Pre-Report: Draft TAR issued for customer review, followed by chamber/bench test correlation (if required).
Final certification-ready TMDR delivered with margin summary and compliance mapping.



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